

SIM1002 Calculus I

Tutorial 4

Derivatives

Exercise 3.2

1. Using the definition, calculate the derivatives of the functions in the following.

i) $F(x) = (x - 1)^2 + 1$; $F'(-1), F'(0), F'(2)$

ii) $k(z) = \frac{1-z}{2z}$; $k'(-1), k'(1), k'(\sqrt{2})$

In the following, find the indicated derivatives by using definition.

2. $\frac{dr}{ds}$ if $r = s^3 - 2s^2 + 3$

3. $\frac{dz}{dw}$ if $z = \frac{1}{\sqrt{w^2-1}}$

In Exercises 4, 5, differentiate the functions. Then find an equation of the tangent line at the indicated point on the graph of the function.

4. $y = f(x) = \frac{8}{\sqrt{x-2}}$, $(x, y) = (6, 4)$

5. $w = g(z) = 1 + \sqrt{4-z}$, $(z, w) = (3, 2)$

Using the alternative Formula for Derivatives.

Use the formula

$$f'(x) = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

6. $g(x) = \frac{x}{x-1}$

7. $g(x) = 1 + \sqrt{x}$